Lect#8:
Project Human Resource Management

What is Project Human Resource Management?

- Project human resource management includes the processes required to make the most effective use of the people involved with a project. Processes include
  - Organizational planning
  - Staff acquisition
  - Team development

Keys to Managing People

- Psychologists and management theorists have devoted much research and thought to the field of managing people at work
- Important areas related to project management include
  - a. motivation
  - b. influence and power
  - c. effectiveness

a. Motivation

- Abraham Maslow developed a hierarchy of needs to illustrate his theory that people's behaviors are guided by a sequence of needs
- Maslow argued that humans possess unique qualities that enable them to make independent choices, thus giving them control of their destiny

Figure 8-1. Maslow's Hierarchy of Needs

b. Influence and power

- Influence
  - Projects are more likely to succeed when project managers influence with
    - expertise
    - work challenge
  - Projects are more likely to fail when project managers rely too heavily on
    - authority
    - money
    - Penalty
- Power
  - the potential ability to influence behavior to get people to do things they would not otherwise do
  - Types of power include
    - Coercive
    - Legitimate
    - Expert
    - Reward
    - Referent

c. Improving Effectiveness - Covey's 7 Habits

- Project managers can apply Covey's 7 habits to improve effectiveness on projects
  - Be proactive
  - Begin with the end in mind
  - Put first things first
  - Think win/win
  - Seek first to understand, then to be understood
  - Synergize
  - Sharpen the saw

Empathic Listening and Rapport

- Good project managers are empathic listeners; they listen with the intent to understand
- Before you can communicate with others, you have to have rapport
- Mirroring is a technique to help establish rapport
- IT professionals often need to develop empathic listening and other people skills to improve relationships with users and other stakeholders

Improving Relationships Between Users and Developers

- Some organizations require business people, not IT people, to take the lead in determining and justifying investments in new computer systems
- CIOs push their staff to recognize that the needs of the business must drive all technology decisions
- Some companies reshape their IT units to look and perform like consulting firms

1. Organizational Planning

- Organizational planning involves identifying, documenting, and assigning project roles, responsibilities, and reporting relationships
- Outputs and processes include
  - project organizational charts
  - work definition and assignment process
  - responsibility assignment matrixes
  - resource histograms

Figure 8-2. Sample Organizational Chart for a Large IT Project

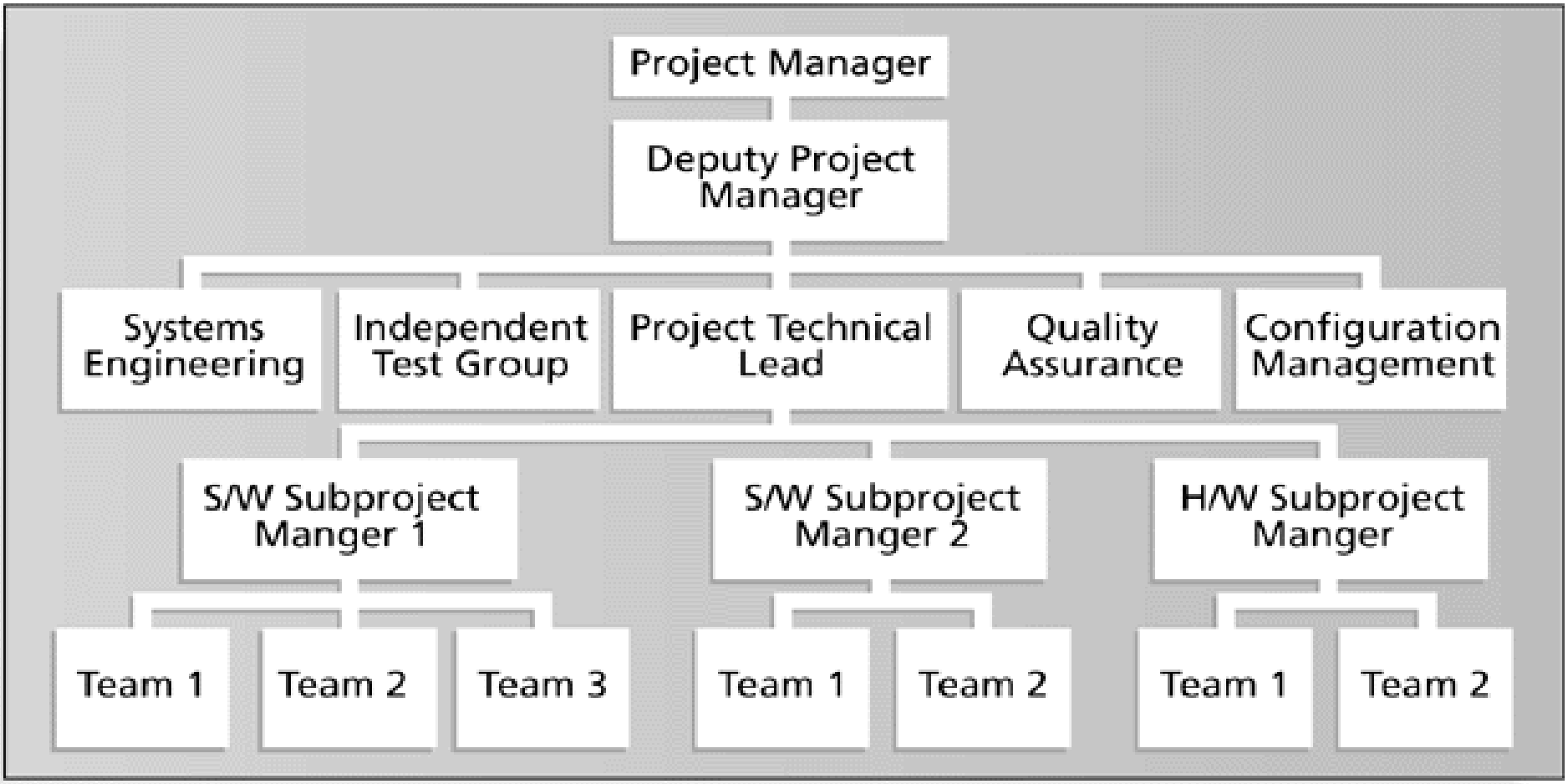


Figure 8-3. Work Definition and Assignment Process

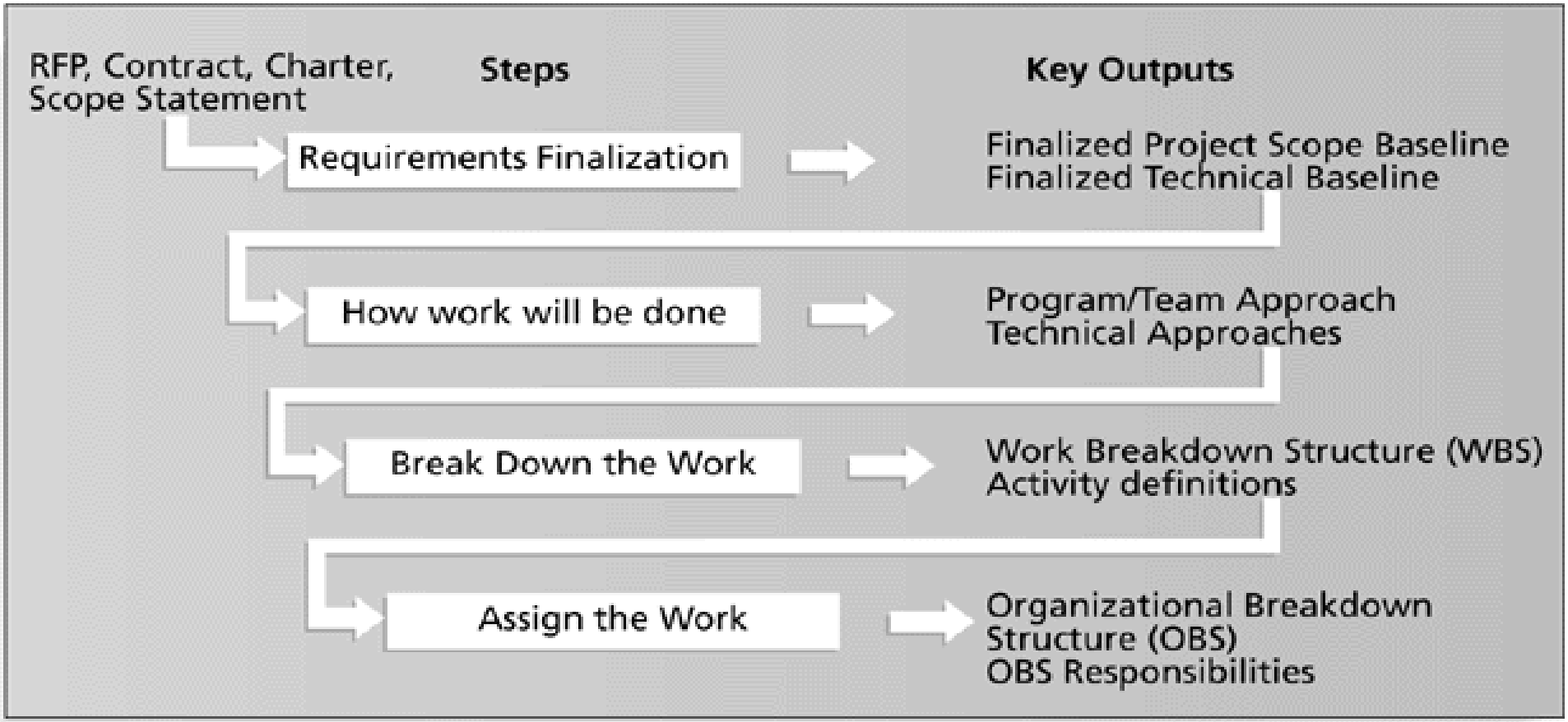


Figure 8-4. Sample Responsibility Assignment Matrix (RAM)

WBS activities →

OBS units ↓

	1.1.1	1.1.2	1.1.3	1.1.4	1.1.5	1.1.6	1.1.7	1.1.8
Systems Engineering	R	R P					R	
Software Development			R P					
Hardware Development				R P				
Test Engineering	P							
Quality Assurance					R P			
Configuration Management						R P		
Integrated Logistics Support							P	
Training								R P

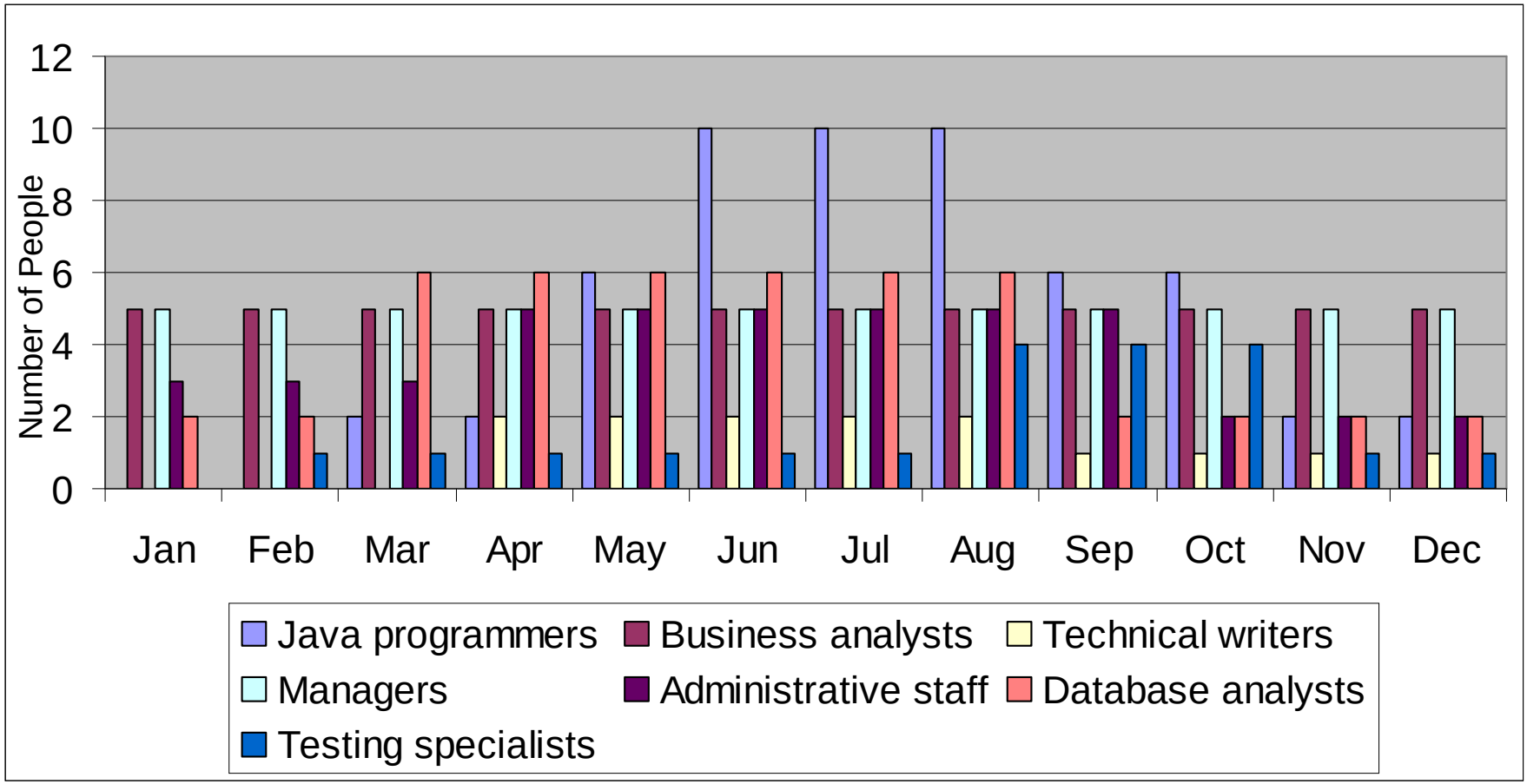
R = Responsible organizational unit
P = Performing organizational unit

Figure 8-5. RAM Showing Stakeholder Roles

	Stakeholders				
Items	A	B	C	D	E
Unit Test	S	A	I	I	R
Integration Test	S	P	A	I	R
System Test	S	P	A	I	R
User Acceptance Test	S	P	I	A	R

A = Accountable
P = Participant
R = Review Required
I = Input Required
S = Sign-off Required

Figure 8-6. Sample Resource Histogram for a Large IT Project



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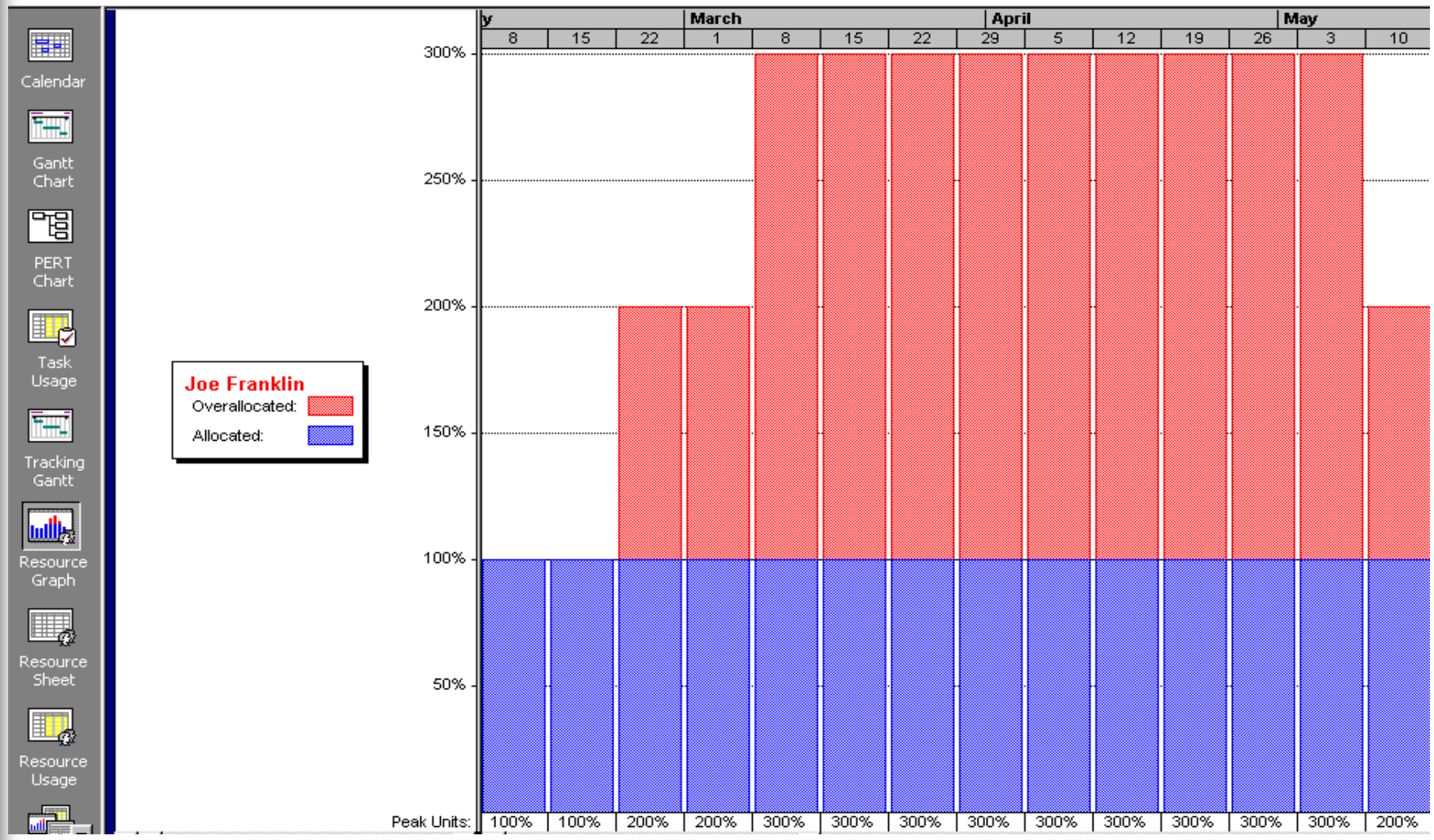
2. Project Staff Acquisition and Team Development

- Staffing plans and good hiring procedures are important in staff acquisition, as are incentives for recruiting and retention
- Some companies give their employees one dollar for every hour a new person they helped hire works
- Some organizations allow people to work from home as an incentive
- Research shows that people leave their jobs because they don’t make a difference, don’t get proper recognition, aren’t learning anything new, don’t like their coworkers, and want to earn more money

Resource Loading and Leveling

- Resource loading refers to the amount of individual resources an existing project schedule requires during specific time periods
- Resource histograms show resource loading
- Over-allocation means more resources than are available are assigned to perform work at a given time

Figure 8-7. Sample Histogram Showing an Over-allocated Individual



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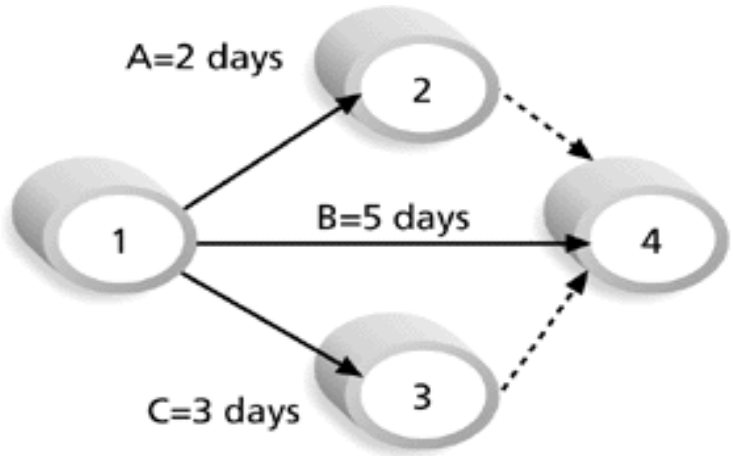
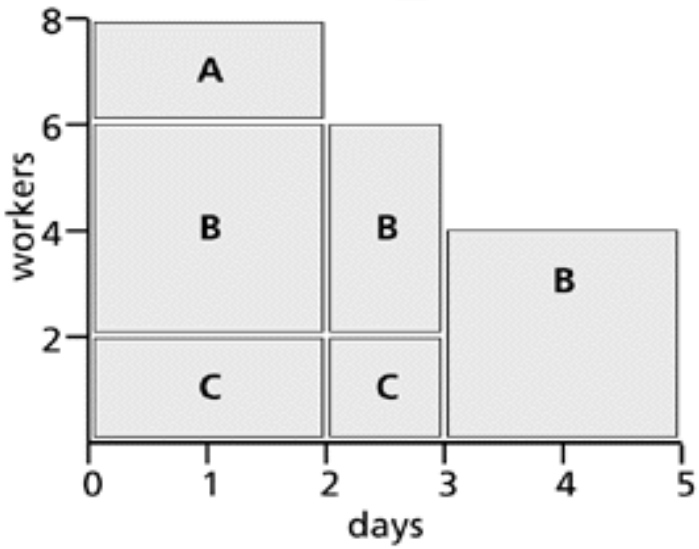
Resource Leveling

- Resource leveling is a technique for resolving resource conflicts by delaying tasks
- The main purpose of resource leveling is to create a smoother distribution of resource usage and reduce over-allocation

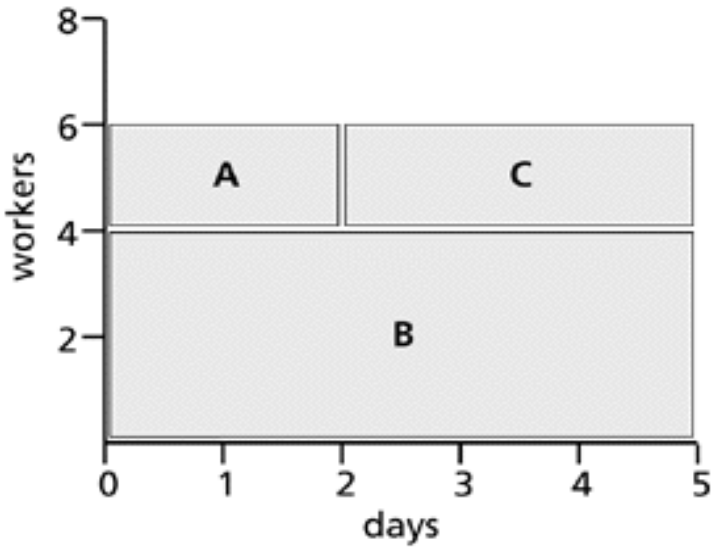
Figure 8-8. Resource Leveling Example



The project network with Activities A, B, and C and durations as shown. Activity A has 3 days of slack, and Activity C has 2 days of slack. Assume Activity A has 2 workers, B has 4 workers, and C has 2 workers.



Resource usage if all Activities start on day one



Resource usage if Activity C is delayed 2 days, its total slack

3. Team Development

- It takes teamwork to successfully complete most projects
- Training can help people understand themselves, each other, and how to work better in teams
- Team building activities include
  - physical challenges
  - psychological preference indicator tools

Meyers-Briggs Type Indicator (MBTI)

- MBTI is a popular tool for determining personality preferences and helping teammates understand each other (www.keirsey.com)
- Four dimensions include:
  - Extrovert/Introvert (E/I)
  - Sensation/Intuition (S/N)
  - Thinking/Feeling (T/F)
  - Judgment/Perception (J/P)
- Most IT professionals are NTs or rationals
- Vary most from general population in not being extroverted or sensing

Reward and Recognition Systems

- Team-based reward and recognition systems can promote teamwork
- Focus on rewarding teams for achieving specific goals
- Allow time for team members to mentor and help each other to meet project goals and develop human resources

General Advice on Teams

- Focus on meeting project objectives and producing positive results
- Fix the problem instead of blaming people
- Establish regular, effective meetings
- Nurture team members and encourage them to help each other
- Acknowledge individual and group accomplishments

Using Software to Assist in Human Resource Management

- Software can help in producing RAMS and resource histograms
- Project management software includes several features related to human resource management such as
  - viewing resource usage information
  - identifying under and over-allocated resources
  - leveling resources